

AutoSampler

Development Changelog & Feature Summary

From an MD coverage sampler to an autonomous, **MSM-convergence-driven** adaptive sampling framework.

This document summarises the new development cycle on the **devel** branch and highlights every new feature and improvement over the original v2.0.0 baseline. All new behaviour is **opt-in**; existing input files keep working unchanged.

Baseline	New (devel)
v2.0.0 — coverage-driven adaptive sampling	MSM-convergence-driven, HPC-scalable, VAMP-2-optimised
Stops on bin-occupancy saturation	Stops on real MSM convergence (timescales / VAMP-2 / error)
Fixed CV / TICA / TVAE / PCA / deep-TICA	+ VAMPNet, SPIB, VAMP-2 feature selection & optimisation
Local multiprocessing only	Local + SLURM + PBS array jobs, fault-tolerant
No tests / CI / docs	96 tests, CI, full docs site, notebook, PDF

Status: 96 automated tests passing; continuous integration green on Python 3.10 & 3.11; ruff-clean. Pull request #1 (devel → main) open.

1. Headline new features

Markov State Model (MSM) convergence engine

The central gap in v2.0.0 — there was no MSM at all, and "convergence" meant bin-occupancy saturation. The new **autosampler/msm/** subsystem (built on deeptime) builds an MSM every iteration and stops sampling when it is genuinely converged.

- **MSMEstimator**: clustering (k-means / regular-space) → transition counts → MLE or Bayesian MSM → implied timescales, VAMP-2 score, PCCA+ metastable states, stationary distribution.
- **ConvergenceMonitor**: composable, pluggable criteria — implied-timescale stability, VAMP-2 plateau, stationary-distribution drift, Bayesian statistical-error thresholds, and a **flux-weighted transition-matrix** criterion (analytic Dirichlet T_{ij} uncertainty) — combined with all / any + patience to require both kinetic resolution and statistical convergence of the transition matrix.
- **MSMSpawner** (spawn_scheme: msm): **uncertainty × leverage × flux** microstate seeding ($\pi_i \cdot |\psi_i| \cdot \sigma_{out,i} + \alpha/\sqrt{c_i}$) that throws runs at the transitions whose in/out rates are uncertain and important; least-counts fallback before the first MSM, with stable clustering for comparable microstate IDs across iterations.

Cutting-edge & optimised collective variables

- New deep CV methods **VAMPNet** and **SPIB** (State Predictive Information Bottleneck) added to a single CV registry beside TICA, TVAE, PCA, deep-TICA, and deep-LDA.
- **VAMP-2 feature selection & optimisation**: automatically select and adaptively update the input features (and even the feature *type*) that best resolve the slow dynamics, via a greedy VAMP-2 optimisation protocol.
- **VAMP-2-driven adaptive retraining**: retrain the CV only when its score on fresh data degrades, instead of on a blind fixed schedule.

Landscape-adaptive binning

The density / weighted-ensemble spawners stratify the CV space into bins. A uniform grid wastes replicas in flat basins and, worse, lets walkers slide back across a wide bin at a barrier before the lag time elapses — stalling the flux. The new **autosampler/binning/adaptive.py** makes bins landscape-adaptive, recomputed every iteration (selected by `binning.scheme`; opt-in, default uniform).

- **gradient**: equi-resistance edges ($\int \exp(\beta F) \propto \int 1/P$) place boundaries where the sampled density is low — fine across barriers, coarse in basins.
- **mab**: Minimal-Adaptive-Binning-style uniform bins between the occupied extremes plus narrow foothold bins at the moving fronts.
- **eigenvector**: bin uniformly along the leading (slowest) CV coordinate only — a committor proxy that is automatically fine at the barrier and folds many CVs into one coordinate.

Scalability: workstation and HPC

- Pluggable execution backends: **local** (multi-GPU workstation), **SLURM**, and **PBS/Torque** — walkers dispatched as scheduler array jobs, one per iteration.
- Fault tolerant: completion is driven by filesystem result markers and failed walkers are automatically resubmitted.
- Switching from a laptop to a CPU-only HPC cluster is a one-line config change.

Analysis, tooling, and end-user experience

- **Weighted-ensemble** resampling: a correct, weight-conserving split/merge implementation (`spawn_scheme: we`), replacing a non-functional placeholder.
- **MSM analysis & plotting**: implied timescales, VAMP-2 / timescale convergence, free-energy surfaces, metastable free energies, and MSM network diagrams, via the `autosampler-analyze` CLI.
- **One input file for everything**: `autosampler-init` writes a fully-annotated YAML exposing every method, feature, and hyperparameter; documented for end users.
- Full documentation site, an executed Jupyter notebook tutorial with rendered plots, and example run scripts for local / SLURM / PBS.

2. Detailed changes by area

Sampling & convergence

- New MSM subsystem (estimator, diagnostics, convergence monitor).
- Flux-weighted transition-matrix convergence criterion (analytic Dirichlet error on T_{ij}) that augments the spectral criteria under mode: all.
- Uncertainty $\times$ leverage $\times$ flux MSM spawner (least-counts fallback) and weighted-ensemble spawner.
- Landscape-adaptive binning (gradient / mab / eigenvector) for the density and weighted-ensemble spawners; opt-in, recomputed each iteration.
- Convergence now based on MSM kinetics, not just spatial coverage (legacy occupancy criterion retained as one selectable option).

Collective variables & features

- VAMPNet and SPIB deep CVs; unified CV method registry with availability checks and actionable install hints.
- VAMP-2 feature scoring, candidate ranking, and greedy column/feature-type optimisation, with adaptive updates during the run.
- Adaptive CV retraining policy driven by the VAMP-2 score.

Execution & scalability

- ExecutionBackend abstraction with local / SLURM / PBS implementations.
- Per-iteration array-job submission, polling, result collection, and automatic resubmission of failed walkers.
- Configurable scheduler resources (partition/queue, walltime, CPUs/GPUs per task, memory, module loads).

Robustness, reproducibility & engineering

- Pydantic v2 configuration with strict validation of every option.
- Checkpoint format versioning; backward-compatible loading of old checkpoints; resume restores MSM, feature-selection, and retraining state.
- MD subprocess timeouts; trajectory-file validation; portable temp paths; narrower exception handling; removed dead code.
- Deterministic seeding across NumPy, PyTorch, and Lightning.
- Test suite (96 tests), GitHub Actions CI, ruff/black/isort, pre-commit, and contribution guidelines.

Analysis & usability

- autosampler-analyze: one-command multi-panel convergence report.
- Matplotlib-free analysis data utilities plus plotting helpers.
- autosampler-init starter input file; full MkDocs documentation; rendered Jupyter notebook tutorial; local/SLURM/PBS example scripts.

3. Capability comparison vs the original

Capability	Original v2.0.0	New (devel)
Convergence	Bin-occupancy saturation	MSM: timescales, VAMP-2, T _{ij} flux-weighted error
MSM building	None	Full pipeline + Bayesian errors + PCCA+
CV methods	fixed, PCA, TICA, TVAE, deep-TICA	+ VAMPNet, SPIB, deep-LDA (unified registry)
Feature choice	Manual, fixed for the run	VAMP-2 selection & optimisation, adaptive
CV retraining	Fixed schedule	Fixed or VAMP-2-adaptive
Spawning	density, voronoi, lof, fps	+ msm (uncertainty×leverage×flux), we (weighted ensemble)
Binning	Uniform grid only	+ gradient / mab / eigenvector (landscape-adaptive)
Execution	Local multiprocessing	Local + SLURM + PBS array jobs, resubmission
Weighted ensemble	Placeholder stub (no-op)	Correct, weight-conserving split/merge
Analysis/plots	None bundled	autosampler-analyze: ITS, VAMP-2, FES, network
Input file	YAML (sparse examples)	autosampler-init annotated template + docs
Tests / CI	None	96 tests, CI (3.10 & 3.11), ruff
Docs / tutorials	README only	MkDocs site, notebook tutorial, changelog PDF

Compatibility: every new capability is opt-in. A v2.0.0 input file runs unchanged; advanced blocks (msm, feature_selection, execution, retrain policy) are simply omitted by default.

4. New configuration options

Every option below is optional; omitting the block keeps the previous behaviour. See the configuration reference and per-topic docs pages.

Block	Key	Default	What it does
msm	convergence_criteria: transition_matrix	—	Flux-weighted Dirichlet uncertainty gate on T_{ij} (params: <code>tol</code> , <code>min_flux</code>).
msm	stable_clustering	false	Seed clustering from previous centres so microstate IDs / T_{ij} stay comparable.
msm	spawn_uncertainty	true	Weight msm spawning by outflow uncertainty ($\times$ leverage $\times$ flux).
msm	spawn_leverage	1	Number of slow eigenvectors used for the leverage factor.
msm	spawn_alpha	1.0	Weight of the least-counts exploration term in msm spawning.
binning	scheme	uniform	uniform gradient mab eigenvector — landscape-adaptive bin placement.
binning	n_fine	100	Density-histogram resolution for the gradient scheme.
binning	smoothing	3	Density smoothing window for the gradient scheme.

Example — drive convergence with the transition-matrix gate and adaptive bins:

```
msm:
  enabled: true
  stable_clustering: true
  convergence_mode: all
  convergence_criteria:
    - {name: vamp2, params: {tol: 0.05}}
    - {name: transition_matrix, params: {tol: 0.2, min_flux: 1.0e-4}}
binning:
  scheme: gradient
```

Generated for the AutoSampler devel branch. See CHANGELOG.md and the documentation for full details.